

Research, innovation and data: a fifth freedom in the EU single market?

Research and innovation should be at the top of the EU economic policy agenda, but cannot over-rely on public investment

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The European Union's single market famously enables four freedoms: the movement of goods, services, capital and labour. Former Italian prime minister Enrico Letta, in his report on the single market issued and discussed by EU leaders in April, proposed a fifth freedom as a top priority, to encompass research, innovation, data and knowledge that have become indispensable drivers of innovation in modern economies (Letta, 2024)¹.

Letta argues that the EU has under-utilised its pools of data, expertise and startups. This wealth of resources benefits global tech giants that are better positioned to capitalise on it and hampers the EU's strategic autonomy and economic security. He claims that it is a necessary extension of the single market for the EU to become a creator of new technologies and foster the development of leading industrial ecosystems of global importance, with a strong European technological infrastructure in areas including data utilisation, artificial intelligence, quantum computing, biotech, bio-robotics and space.

The Letta report contains a number of constructive and innovative ideas. Most importantly, he does not just attempt to put the fifth freedom on par with other single market freedoms. Instead, he puts it squarely at the top of all single market freedoms: innovation as the necessary condition for the success of all other freedoms, indeed for the success of the EU as an economic project.

Should this be accepted and effectively implemented by the next European Commission, it could herald a major shift in the EU policy environment, which often prioritises precaution over innovation. It would also be a recognition that the empirical evidence of a slow-down in EU productivity growth, and thus in innovation, should be taken seriously (see for example Pinkus *et al*, 2024). EU productivity growth since the 2009 financial crisis has lagged about a third behind the US. That undermines the EU's long-term economic welfare.

On the other hand, the fifth freedom sits somewhat uncomfortably in a report on the single market because it has little to do with geographical obstacles or borders. Insisting on the freedom to investigate, explore and create in a borderless single market feels like pushing at an open door. There are hardly any EU internal borders to the mobility of research projects, knowledge and researchers.

Helping data flow

Another positive message from the report is that digital data assumes a central role in Letta's view of the knowledge economy. Data is a new production factor in modern economies. Eliminating barriers to data access is a powerful catalyst for innovation. Access to computing power and AI technologies are also a necessary ingredient. Letta acknowledges that considerable progress has already been made with several EU digital laws, including the Digital Markets Act, the Digital Services Act, the Data Act and the Data Governance Act. But these are considered insufficient to nurture the necessary level of innovation.

Letta supports the development of European data spaces in key sectors, in line with the European Commission's (2020) data strategy. Opening access to data and creating data pooling spaces can leverage the value of digital data as a new production factor. Data portability is beneficial because data, once collected by one party for a particular purpose, can often be re-used by other parties for competing or complementary purposes. Data portability thus stimulates competition and innovation in data-driven services markets. Data pooling generates other types of benefits. The valuable insights that can be extracted from a large data pool often exceed the insights that can be extracted from fragmented and smaller datasets.

Letta refers to healthcare as an example of the implementation of the fifth freedom. He cites the European Health Data Space (EHDS) regulation that facilitates the portability

of personal health data between medical service providers to stimulate competition between services, and will also create an EU-wide health data pool for research purposes to stimulate medical innovation². EHDS would have been a good template for other data pooling initiatives. Unfortunately, other sectoral data pools may not be as generous to researchers and innovation. Preliminary ideas for a Common European Agricultural Data Space emphasise exclusive data rights for farms, at the expense of data pooling for innovation purposes³.

Typically, Letta's report recommends removing barriers to cross-border data flows by means of interoperability and data regulations. But there are few restrictions on cross-border data flows inside the EU⁴. The real obstacles to access to data are located inside firms that collect and store data in their proprietary silos. They are reluctant to share data with users who generated the data or third parties selected by users, let alone in a common data pool accessible to many users.

The EU Data Act (Regulation (EU) 2023/2854) also makes data pooling difficult. It attributes exclusive data licensing and monopolistic data-pricing rights to device manufacturers, restricting data access for users to very narrowly defined datasets and limiting the use of this data for competitive purposes. That slows down data-driven innovation. The EU Data Governance Act (Regulation (EU) 2022/868) meanwhile does not drive innovation because it excludes precisely those platforms that produce data-driven innovation services, including data analytics, transformation and extraction of value-added from data pools. Over the last couple of years, EU data policies have moved back and forth between the debunked concept of private data ownership and the recognition that data sharing is beneficial for innovation and competition. If the European Commission is to take Letta seriously, it should move away decisively from exclusive data rights and start to see data as a collectively generated production factor that should be leveraged as a major driver of innovation in the digital economy.

Redistributing rents

The EU Digital Markets Act (DMA, Regulation (EU) 2022/1925) is a pioneering attempt to weaken the monopolistic market power of mostly US-based 'gatekeeper' platforms. If implemented correctly, it could redistribute some of these monopoly rents to EU consumers and small businesses. But will this redistribution evaporate in non-investable consumer surplus and fragmented financial resources? The EU will need financial instruments to channel these resources back into digital R&D and innovative start-up capital. Even after implementation of the DMA, Europe's advanced digital

technologies may still rely on US platforms to bring their services to consumers and businesses.

It is unlikely that EU publicly financed R&D can compete with these platforms, which are all privately financed for-profit companies. Letta recognises the need to mobilise more private investment as a complement to public-sector investment. One of his most interesting proposals is the creation of an *“EU Stock Exchange for Deep Tech”* companies that use cutting-edge science and technology, including AI, quantum and biotechnology. Start-ups are high-risk undertakings but offer high gains, if successful. In the EU, because of banking regulations, these types of risky asset are downgraded. The EU should facilitate the creation of a deep-tech stock exchange with specific rules adapted to this risk class.

Letta also argues in favour of creating a strong digital infrastructure layer through consolidation in the telecom sector, allowing many small national telecom providers to merge cross-border into a dozen or so large providers that can invest in advanced infrastructures, including 5G and 6G mobile networks. Among his more provocative and promising ideas, Letta suggests that the time may have come to re-evaluate rules on net neutrality, or non-discriminatory treatment of online traffic. Different treatment of different types of data flows allows optimisation of connectivity, which is important for robotics, the internet of things and AI. It would allow the introduction of innovative use cases that are currently non-compliant with net neutrality. In the US, the FTC abandoned net neutrality in 2018, without major upheaval in the sector.

Another important infrastructure component is cloud-computing capacity for the development of AI models, in which Letta endorses EU public investment. The EU is running far behind the US platforms, which invest massively in cloud computing. Letta suggests that the EU should prioritise shared networks of computational resources and supercomputers, such as the EU's High-Performance Computing (HPC) initiative⁵. Unfortunately, HPC's centralised and public sector governance model is more adapted to academic research than to the requirements of AI start-ups, which require flexible and scalable computing capacity and dedicated AI processors. The public sector will not be able to match the financial resources of the big platforms to invest in hyperscale computing capacity for AI models.

Unfortunately, some of Letta's more concrete implementation proposals boil down to old European wines in barely new bottles. Letta proposes the creation of a European Knowledge Commons, a centralised digital platform to provide access to publicly

funded research, datasets and educational resources, allowing citizens and businesses to tap into a wealth of knowledge for innovation. This Commons should be accompanied by facilitation of cross-border data flows, development of European data spaces, creation of data regulation sandboxes and promotion of researcher mobility within the European Research Area and efforts to retain talent in Europe. There is nothing new in these proposals. Access to EU research findings and data has already opened up considerably over the last decade. But Letta does not mention how sharing knowledge in a commons can be squared with incentives to invest in patentable and commercially exploitable research. Public-private partnerships in strategic areas focused on knowledge exchange and innovation uptake may be important at the research stage, but often become more problematic at the commercial stage, when the public sector may need to spin off successful projects to the private sector.

Almost inevitably, Europe's old ideas about selecting industrial champions pop up again. Letta clearly favours a centralist, public-sector-driven approach to innovation, in order to be able to draw in substantial private investments. He emphasises public sector financing and commons-based approaches to knowledge accumulation and innovation. He claims that establishing European technological infrastructure involves granting authority to a collective industrial policy at European scale, moving beyond national confines. This is unfortunate. One would have expected the Letta report to acknowledge the enormous private-sector contribution to innovation and productivity growth.

AI and cybersecurity

Letta underscores the importance of the development and deployment of AI technologies, including ethical guidelines and regulatory compliance standards. He argues that even if the most powerful AI models have been developed outside the EU, it can still win the race to make the most of AI applications. He expresses belief in the EU's position as a leading hub for AI innovation.

This optimistic view may be difficult to square with the realities of the EU AI Act that will impose considerable compliance costs on smaller EU AI developers and may complicate access to model training data⁶. Combined with the lack of adequate computing capacity and finance in the EU, it is easy to understand why EU AI start-ups have opted for collaboration with US platforms. That collaboration gives them access to computing power and inputs data, and also to commercial outlets for their AI model applications, a prerequisite for generating revenue. It is difficult for a start-up to launch

a new business model from scratch. Smaller models and applications on top of existing foundation models may have some commercial future but they are vulnerable to competition and high intermediation fees from Big Tech AI services.

Letta's report rightly observes that the current fragmentation in cybersecurity standards hampers the development of robust security capabilities, by preventing network operators from leveraging centralised network architectures that could benefit from economies of scale. Fragmented national cybersecurity standards and reporting requirements undermine the efficiency of cybersecurity strategies at EU level. Much cybersecurity work is done by giants such as Google and Microsoft because they have a global overview of threats through their sprawling consumer- and business-facing networks. In the absence of EU players of this size, and further handicapped by fragmented national regulation, this remains a source of concern, especially at a time when cyberwarfare is increasingly important.

In sum, the main merit of Letta's fifth freedom idea is that it puts research and innovation back at the top of the EU economic policy agenda, to counter the slow-down in EU productivity growth. Wrapping it up in a single market freedom anchors it very well in the EU's institutional and policy foundations. Bringing in data and AI policies to leverage productivity growth chimes with current frontier technologies. Several of his policy proposals challenge the current status quo. However, his reliance on public sector-led industrial policies and investment ignores the fact that private sector R&D and investment now vastly exceed public-sector financing capacities. A Knowledge Commons overlooks the fact that private appropriation of innovation rents has become the main driver of R&D financing. Letta's ideas will require considerable polishing and fine-tuning to make them fit in the realities of today's innovation economics.

References

Delors, J. (1989) 'Address given by Jacques Delors to the European Parliament (17 January 1989)', available at https://www.cvce.eu/obj/address_given_by_jacques_delors_to_the_european_parliament_17_january_1989-en-b9c06b95-db97-4774-a700-e8aea5172233.html

European Commission (2017) 'Impact assessment accompanying proposal for a Regulation on a framework for the free flow of non-personal data in the European

Union', SWD(2017) 304 final, available at <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=SWD:2017:304:FIN>

European Commission (2020) 'A European strategy for data', COM(2020) 66 final, available at <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52020DC0066>

Letta, E. (2024) *Much More Than a Market*, report to the European Council, available at <https://www.consilium.europa.eu/media/ny3j24sm/much-more-than-a-market-report-by-enrico-letta.pdf>

Pinkus, D., J. Pisani-Ferry, S. Tagliapietra, R. Veugelers, G. Zachmann and J. Zettelmeyer (2024) *Coordination for competitiveness*, study requested by the ECON committee, European Parliament, available at [https://www.europarl.europa.eu/RegData/etudes/STUD/2024/747838/IPOL_STU\(2024\)747838_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/STUD/2024/747838/IPOL_STU(2024)747838_EN.pdf)

Endnotes

- 1 The idea of a fifth freedom is not new. It was mentioned in 1989 by then European Commission president Jacques Delors (Delors, 1989), and by former science and research commissioner Janez Potocnik in 2007. See European Commission archived content, 'Make "knowledge" a fifth Community freedom, says Potocnik at Green Paper launch', [https://cordis.europa.eu/article/id/27454-make-knowledge-a-fifth-commun....](https://cordis.europa.eu/article/id/27454-make-knowledge-a-fifth-commun...)
- 2 At time of writing, EHDS has been agreed but not fully ratified; see European Commission press release of 24 April 2024, 'Commission welcomes European Parliament's adoption of the European Health Data Space and regulation on substances of human origin', https://ec.europa.eu/commission/presscorner/detail/en/IP_24_2250.
- 3 See the AgriDataSpace project, <https://agridataspace-csa.eu/>.
- 4 Annex 5 in European Commission (2017) detected some restrictions, mostly on administrative and tax data, that represent only a tiny part of total data flows.
- 5 See https://eurohpc-ju.europa.eu/index_en.
- 6 The Act has been approved but not published in its final form. See Council of the EU press release of 21 May 2024, 'Artificial intelligence (AI) act: Council gives final green light to the first worldwide rules on AI', <https://www.consilium.europa.eu/en/press/press-releases/2024/05/21/arti...>

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